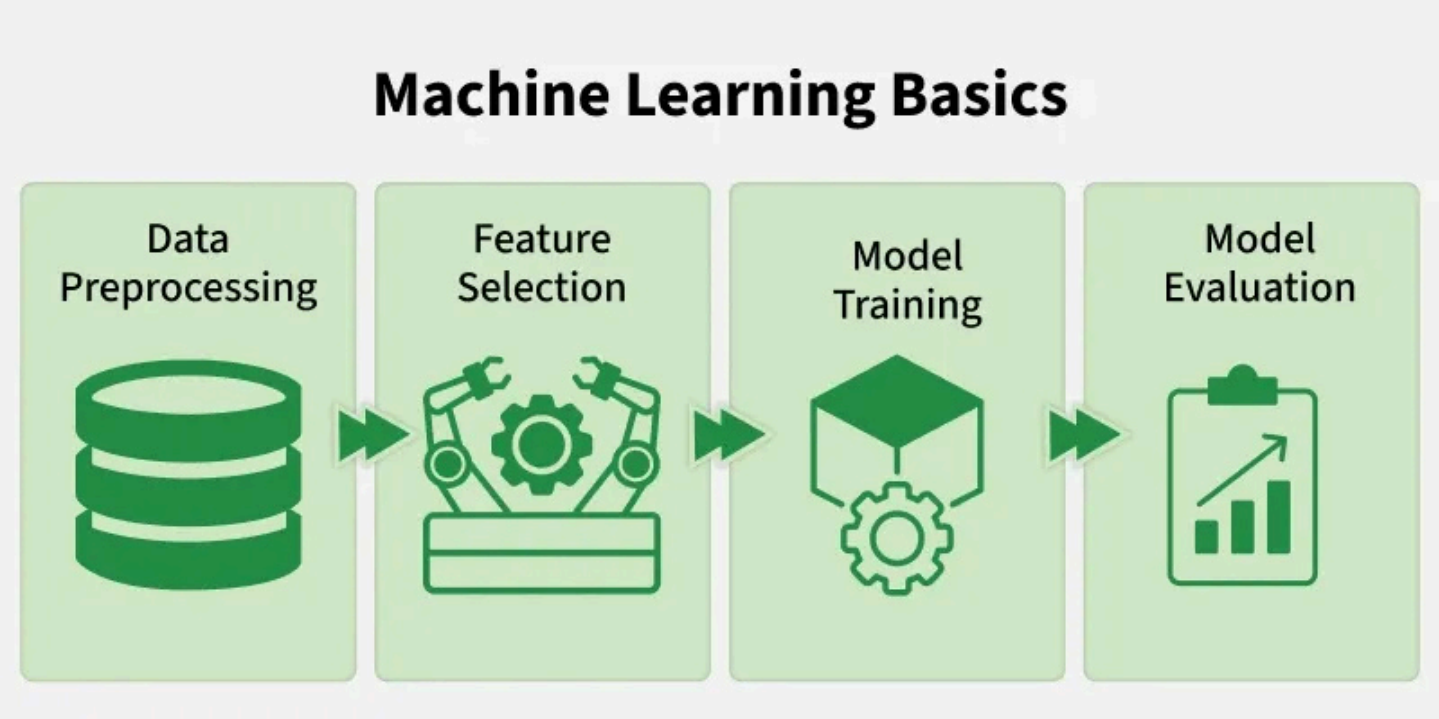
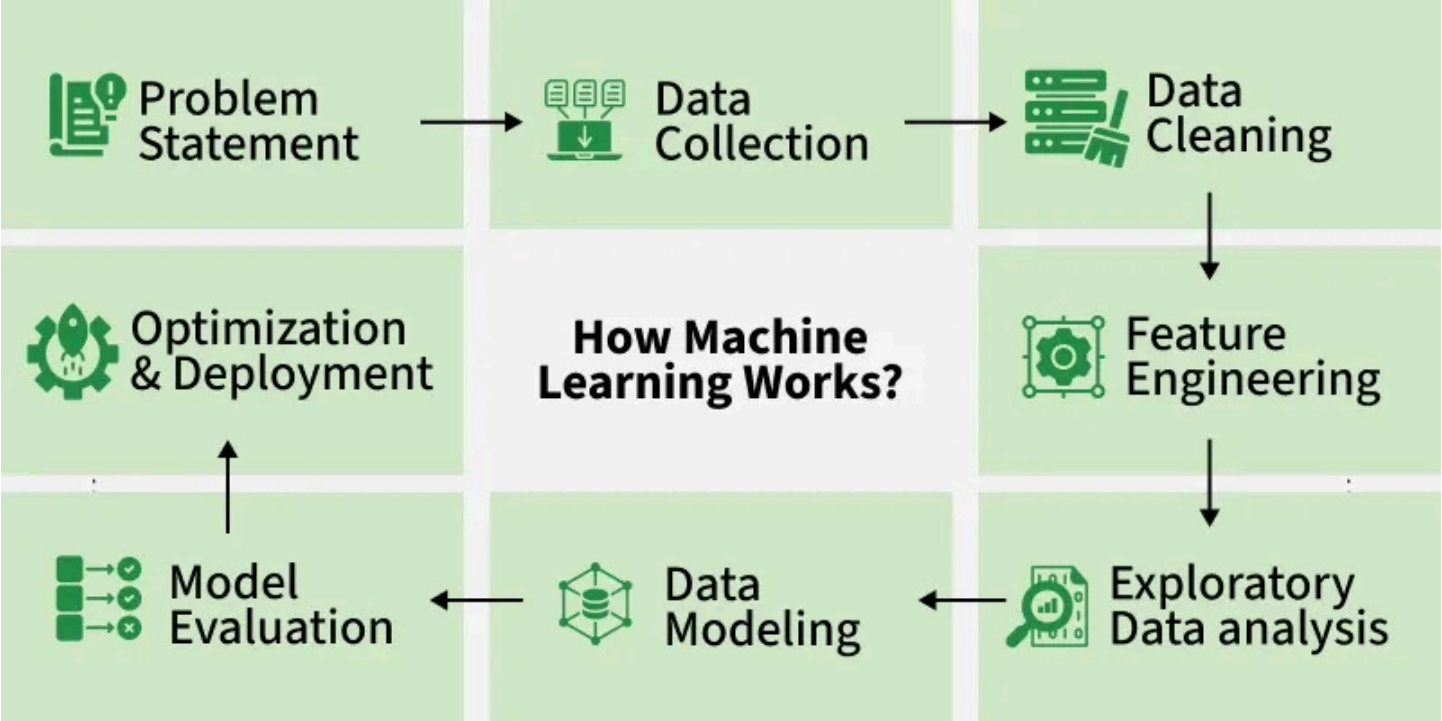
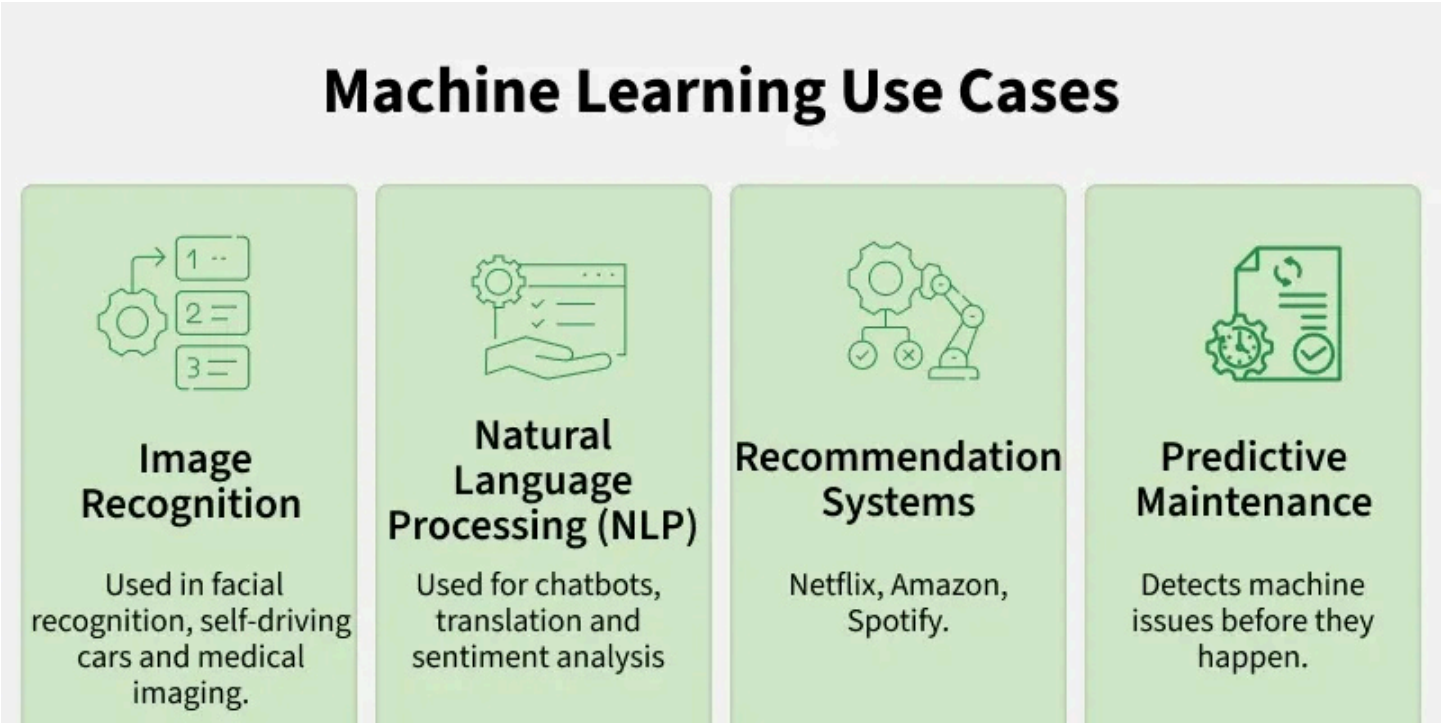
MACHINE LEARNING

What is Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables systems to automatically learn and improve from experience without being explicitly programmed. It focuses on developing algorithms that can analyze data, identify patterns, and make decisions.







Types

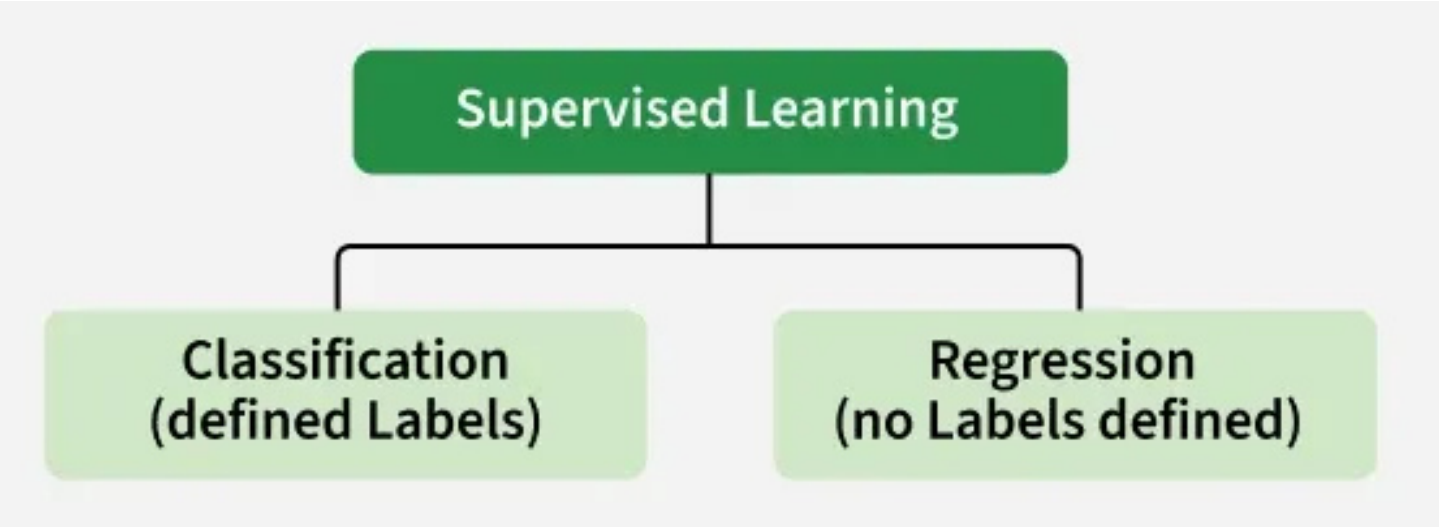
Machine Learning is mainly divided into three core types:

- **Supervised Learning:** Supervised learning is a type of machine learning where a model learns from labelled data, meaning each input has a correct output. The model compares its predictions with actual results and improves over time to increase accuracy.

Its main features are:

- Each input has a known output
- Adjusts itself to reduce prediction errors
- Make accurate predictions on new data
- For example it recognizes handwritten digits from trained data
- Each input has a known

Types of Supervised Learning



Classification: Where the output is a categorical variable (e.g., spam vs. non-spam emails, yes vs. no).

Regression: Where the output is a continuous variable (e.g., predicting house prices, stock prices).

- ▢ **Unsupervised Learning:** Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.
- It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.
- Helps identify hidden patterns in data
- Useful for grouping, compression and anomaly detection

- ▢ **Reinforcement Learning**: Learns through trial and error to maximize rewards, ideal for decision-making tasks. Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal. The agent performs actions and receives feedback to optimize its decision-making over time.
- Agent: The decision-maker that performs actions.
- Environment: The world or system in which the agent operates.
- State: The situation or condition the agent is currently in.
- Action: The possible moves or decisions the agent can make.
- Reward: The feedback or result from the environment based on the agent's action.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression, which predicts continuous values it predicts the probability that an input belongs to a specific class.

- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs into a probability value between 0 and 1.

Types of Logistic Regression

Logistic regression can be classified into three main types based on the nature of the dependent variable:

1. **Binomial Logistic Regression**: This type is used when the dependent variable has only two possible categories. Examples include Yes/No, Pass/Fail or 0/1. It is the most common form of logistic regression and is used for binary classification problems.
2. **Multinomial Logistic Regression**: This is used when the dependent variable has three or more possible categories that are not ordered. For example, classifying animals into categories like "cat," "dog" or "sheep." It extends the binary logistic regression to handle multiple classes.
3. **Ordinal Logistic Regression**: This type applies when the dependent variable has three or more categories with a natural order or ranking. Examples include ratings

like "low," "medium" and "high." It takes the order of the categories into account when modeling

Linear Regression

Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.

- It assumes that there is a linear relationship between the input and output
- Uses a best-fit line to make predictions
- Commonly used in forecasting, trend analysis, and predictive modelling

Ensemble Learning

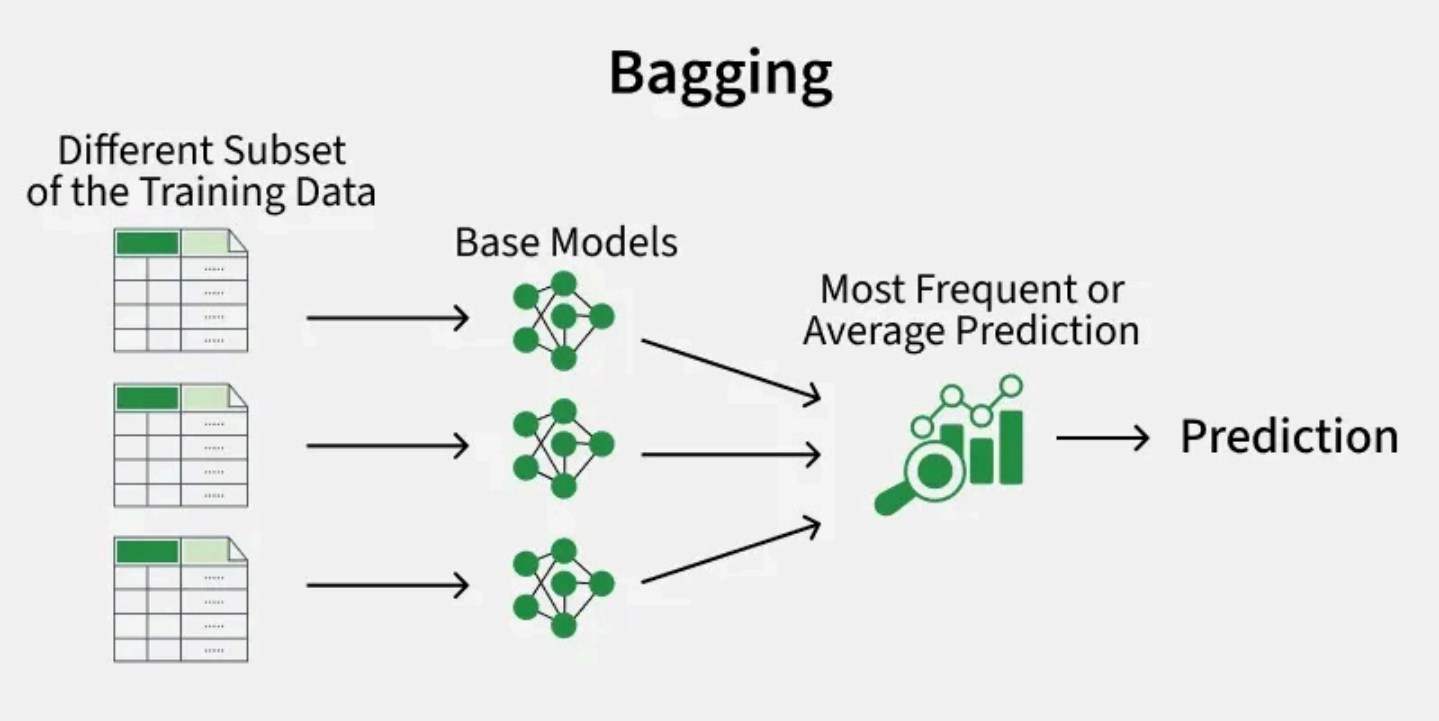
Ensemble learning is a method where multiple models are combined instead of using just one. Even if individual models are weak, combining their results gives more accurate and reliable predictions.

- Uses multiple models together to improve overall accuracy.
- Reduces errors by balancing mistakes across models.
- Works on a simple idea similar to combining opinions from a group

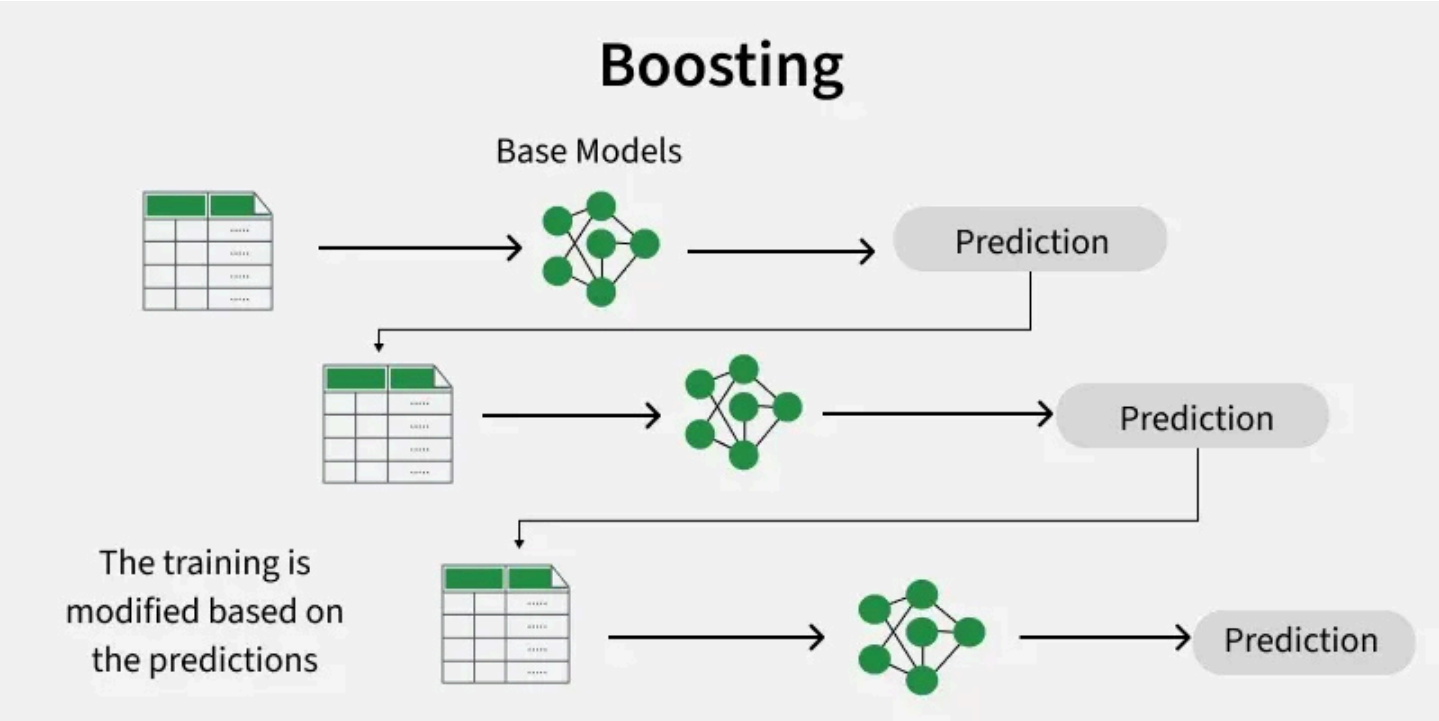
Types of Ensemble Learning

There are three main types of ensemble methods:

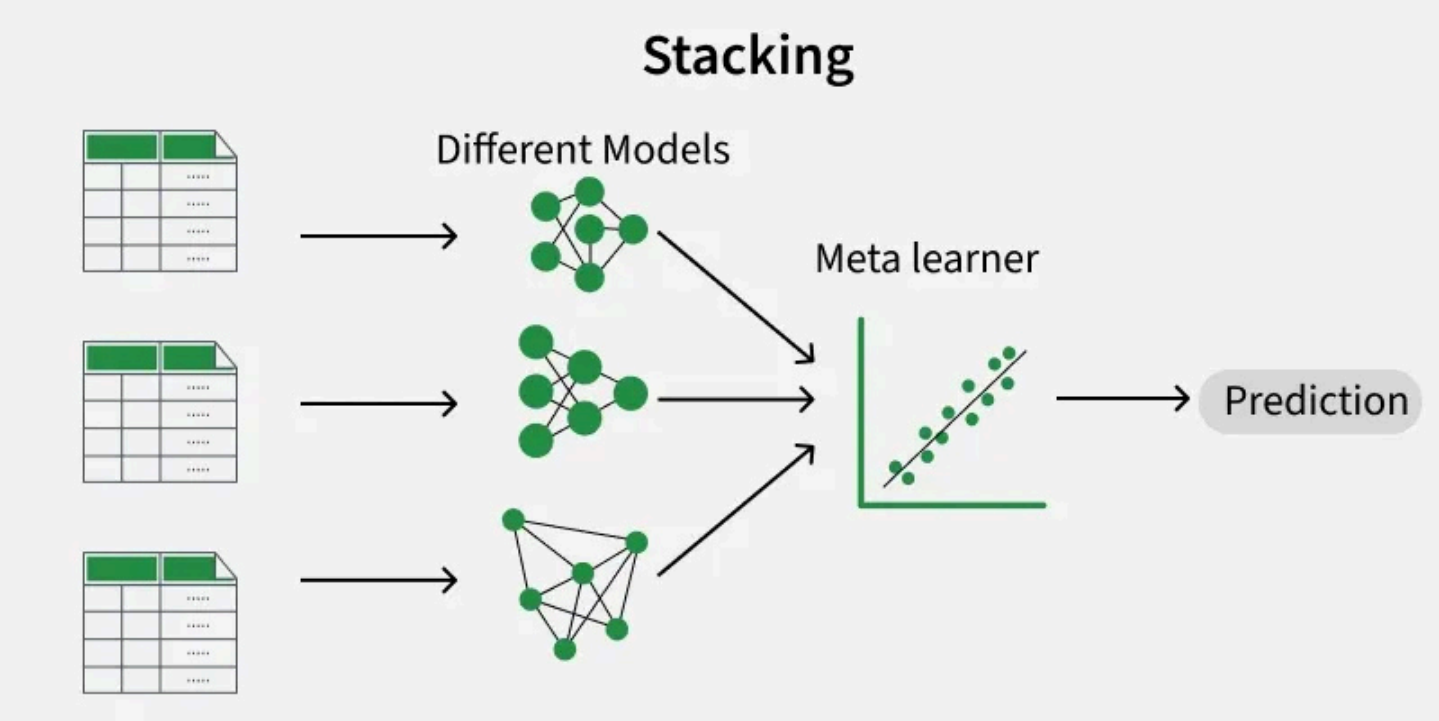
1. **Bagging (Bootstrap Aggregating):** Models are trained independently on different random subsets of the training data. Their results are then combined—usually by averaging (for regression) or voting (for classification). This helps reduce variance and prevents overfitting.



1. **Boosting:** Models are trained one after another. Each new model focuses on fixing the errors made by the previous ones. The final prediction is a weighted combination of all models, which helps reduce bias and improve accuracy.



1. **Stacking (Stacked Generalization):** Multiple different models (often of different types) are trained and their predictions are used as inputs to a final model, called a meta-model. The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.



Unsupervised Machine Learning

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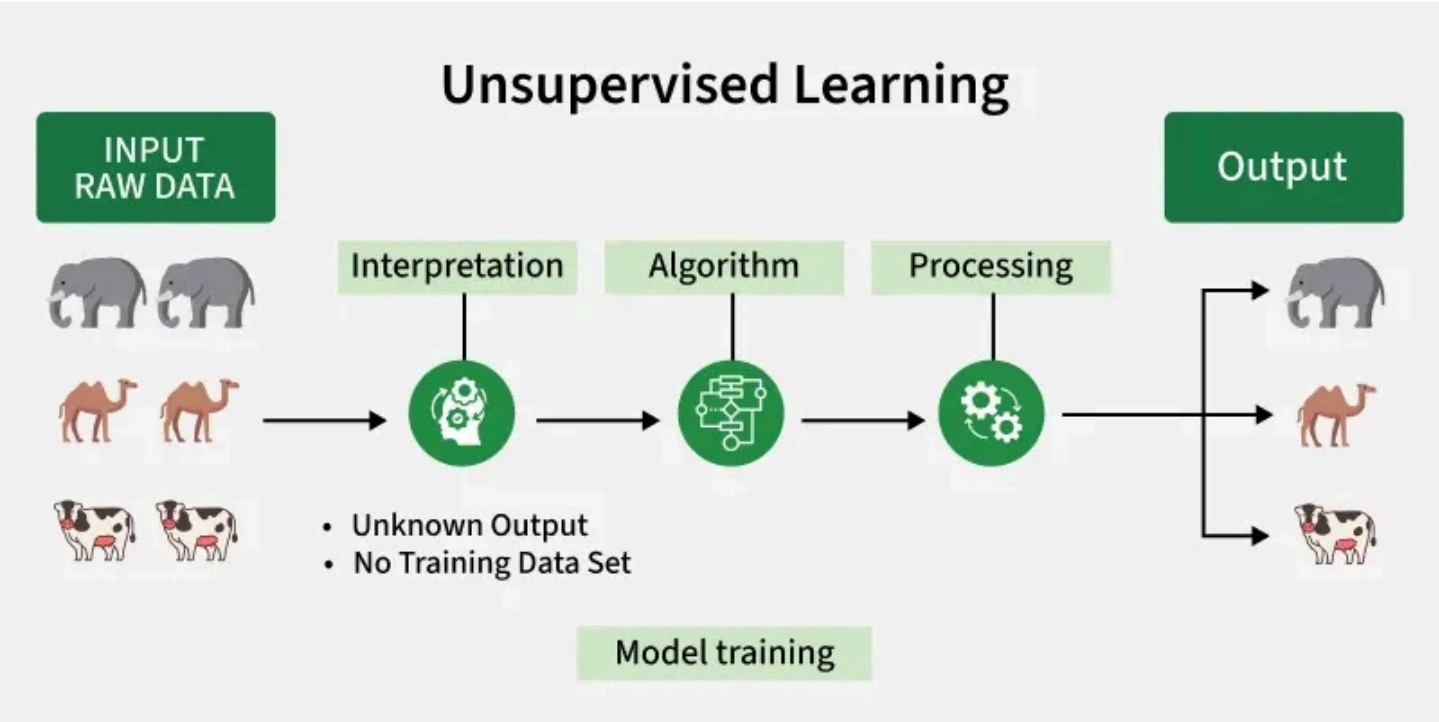

Unsupervised Learning Algorithms

There are mainly 3 types of Unsupervised Algorithms that are used:

1. Clustering Algorithms

Clustering is an unsupervised machine learning technique that groups unlabeled data into clusters based on similarity. Its goal is to discover patterns or relationships within the data without any prior knowledge of categories or labels.

- Groups data points that share similar features or characteristics.
- Helps find natural groupings in raw, unclassified data.
- Commonly used for customer segmentation, anomaly detection and data organization.
- Works purely from the input data without any output labels.
- Enables understanding of data structure for further analysis or decision-making.

2. Association Rule Learning

Association rule learning is a rule-based unsupervised learning technique used to discover interesting relationships between variables in large datasets. It identifies patterns in the form of “if-then” rules, showing how the presence of some items in the data implies the presence of others.

- Finds frequent item combinations and the rules connecting them.
- Commonly used in market basket analysis to understand product purchase relationships.
- Helps retailers design promotions and cross-selling strategies.

3. Dimensionality Reduction

Dimensionality reduction is the process of decreasing the number of features or variables in a dataset while retaining as much of the original information as possible. This technique helps simplify complex data making it easier to analyze and visualize. It also improves the efficiency and performance of machine learning algorithms by reducing noise and computational cost.

- It reduces the dataset’s feature space from many dimensions to fewer, more meaningful ones.
- Helps focus on the most important traits or patterns in the data.
- Commonly used to improve model speed and reduce overfitting.

https://github.com/gourisatheesh00-web/AI-internship